CSE 21 Fall 2026
Homework 4
Due date: Monday, October 26 at 11:59pm

1. (12 points) Let $(a_1, \ldots, a_n)$ be a random permutation of the elements $\{1, \ldots, n\}$ selected uniformly at random. Compute each expected value using linearity of expectation:

   (a) Compute the expected number of indices that have the property that $i < a_i$.

   (b) Compute the expected number of interior elements a_i such that $a_{i-1} < a_i < a_{i+1}$.

   (c) Compute the expected number of swaps (A swap is a pair of elements (a_i, a_j) such that $a_i = j$ and $a_j = i$ and $i < j$.)

2. (16 points)

   (a) An art museum with 12 different rooms just got a shipment of 9 new *different* paintings. The museum curator puts the paintings in the rooms uniformly at random (each of the 12^9 arrangements is equally likely.)

      i. Compute the expected number of rooms without any paintings.

      ii. Compute the expected number of rooms with exactly 1 painting.

   (b) An art museum with 12 different rooms just got a shipment of 9 *indistinguishable* prints. The museum curator puts the prints in the rooms uniformly at random (each of the $\binom{9+12-1}{12-1}$ arrangements is equally likely.)

      i. Compute the expected number of rooms without any prints.

      ii. Compute the expected number of rooms with exactly 1 print.

3. (8 points)

   (a) What is the expected number of coin flips of a fair coin until you get two consecutive H's?
   (Hint: use the law of total expectation and use algebra to solve the equation:

   $$E(X) = E(X|T)P(T) + E(X|HT)P(HT) + E(X|HH)P(HH)$$

   where X is the random variable of the number of flips until you get two consecutive H's, T is the event that your first flip is tails, HT is the event that your first two flips are heads, tails, and HH is the event that the first two flips are HH.)

   (b) What is the expected number of coin flips until you get a HHH?
   (You can use a similar strategy for this problem)

4. (8 points) An ideal hash function from $\{1, \ldots, m\}$ to $\{1, \ldots, n\}$ is a function that is selected uniformly at random from the set of all functions that have domain $\{1, \ldots, m\}$ and codomain $\{1, \ldots, n\}$.

   (a) Let H be an ideal hash function from $\{1, \ldots, m\}$ to $\{1, \ldots, n\}$. What is the expected number of collisions? (a collision is a pair of domain elements $i < j$ such that $H(i) = H(j)$.

   (b) Let H be an ideal hash function from $\{1, \ldots, m\}$ to $\{1, \ldots, n\}$. What is the expected number of codomain values that do not get mapped to?